



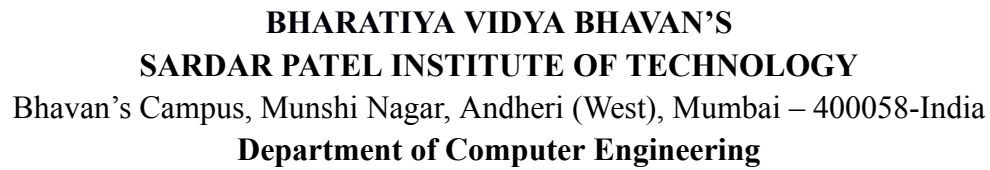
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Experiment No.	Lab7 Packet Crafting

TASK 1

Packet Crafted

```
[>>> packet = IP(dst="8.8.8.8")/ICMP()
[>>> packet.show()
###[ IP ]###
  version    = 4
  ihl        = None
  tos        = 0x0
  len        = None
  id         = 1
  flags      =
  frag       = 0
  ttl        = 64
  proto      = icmp
  checksum   = None
  src        = 10.10.72.112
  dst        = 8.8.8.8
  \options   \
###[ ICMP ]###
  type       = echo-request
  code       = 0
  checksum   = None
  id         = 0x0
  seq        = 0x0
  unused     = b''
```



```
[>>> reply = sr1(packet)
Begin emission
...
Finished sending 1 packets
*
Received 4 packets, got 1 answers, remaining 0 packets
[>>> reply.show()
#### [ IP ] ####
version      = 4
ihl          = 5
tos          = 0x0
len          = 28
id           = 0
flags        = 
frag         = 0
ttl          = 116
proto        = icmp
chksum       = 0xe457
src          = 8.8.8.8
dst          = 10.10.72.112
\options     \
#### [ ICMP ] ####
type         = echo-reply
code         = 0
chksum       = 0x0
id           = 0x0
seq          = 0x0
unused       = b''
#### [ Padding ] ####
load         = b'\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'
```

Capturing from Wi-Fi: en0

icmp

No.	Time	Source	Destination	Protocol	Length	Info
5267	187.537534	10.10.72.112	142.250.207.227	ICMP	70	Destination unreachable (Port unreachable)
5504	114.279555	10.10.72.112	142.250.207.227	ICMP	70	Destination unreachable (Port unreachable)
6391	132.394503	10.10.72.112	142.250.207.227	ICMP	70	Destination unreachable (Port unreachable)
6493	136.705403	10.10.72.112	142.250.207.227	ICMP	70	Destination unreachable (Port unreachable)
7604	145.205029	10.10.72.112	142.250.207.227	ICMP	70	Destination unreachable (Port unreachable)
8150	150.017759	10.10.72.112	172.64.144.52	ICMP	70	Destination unreachable (Port unreachable)
8155	151.150221	10.10.72.112	142.250.207.227	ICMP	70	Destination unreachable (Port unreachable)
8454	155.505388	10.10.72.112	172.64.144.52	ICMP	70	Destination unreachable (Port unreachable)
9461	179.975173	10.10.72.112	8.8.8.8	ICMP	42	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 9462)
9462	179.981742	8.8.8.8	10.10.72.112	ICMP	60	Echo (ping) reply id=0x0000, seq=0/0, ttl=116 (request in 9461)
13013	273.375747	10.10.72.112	172.64.155.209	ICMP	70	Destination unreachable (Port unreachable)
13018	273.417877	10.10.72.112	104.18.32.47	ICMP	70	Destination unreachable (Port unreachable)
13021	273.424554	10.10.72.112	172.64.155.209	ICMP	70	Destination unreachable (Port unreachable)

> Frame 9461: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface en0, id 0
 > Ethernet II, Src: 92:a4:01:a5:59:52 (92:a4:01:a5:59:52), Dst: CiscoMeraki_4f:00:01 (00:18:0a:4f:00:01)
 > Internet Protocol Version 4, Src: 10.10.72.112, Dst: 8.8.8.8
 > Internet Control Message Protocol

Internet Control Message Protocol: Protocol

Packets: 19452 - Displayed: 36 (0.2%)

Profile: Default



TASK 2

Packet Crafted

```
[>>> packet = IP(dst="8.8.8.8")/UDP(dport=53)/"Hello"
[>>> reply = sr1(packet, timeout=2)
Begin emission
...
Finished sending 1 packets
.....
Received 28 packets, got 0 answers, remaining 1 packets
[>>> packet = IP(dst="8.8.8.8")/UDP(dport=53)/"Hello"
[>>> packet.show()
#### [ IP ] ####
  version   = 4
  ihl       = None
  tos       = 0x0
  len       = None
  id        = 1
  flags     =
  frag      = 0
  ttl       = 64
  proto     = udp
  chksum    = None
  src       = 10.10.72.112
  dst       = 8.8.8.8
  \options  \
#### [ UDP ] ####
  sport      = domain
  dport      = domain
  len        = None
  chksum     = None
#### [ Raw ] ####
  load       = b'Hello'
```



Packet Response

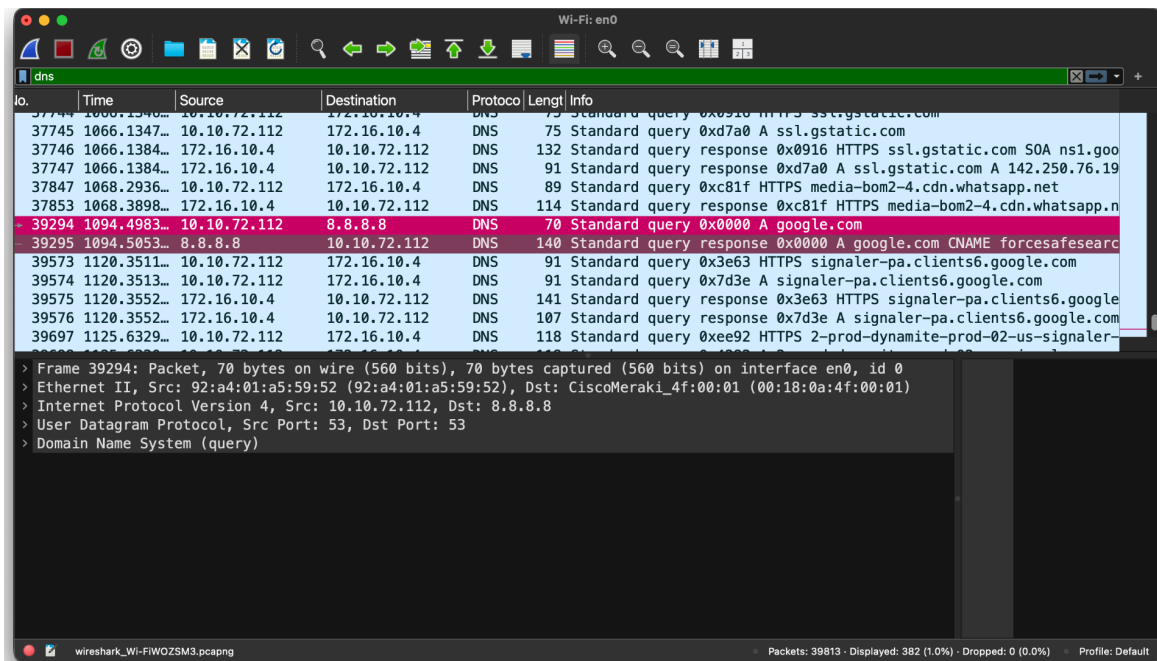
```
[>>> reply = sr1(packet, timeout=2)
Begin emission
...
Finished sending 1 packets
.....
Received 15 packets, got 0 answers, remaining 1 packets
[>>> reply.show()

-----
AttributeError                                Traceback (most recent call last)
Cell In[10], line 1
----> 1 reply.show()

AttributeError: 'NoneType' object has no attribute 'show'
```

Dns Query

```
[>>> packet = IP(dst="8.8.8.8")/UDP(dport=53)/DNS(rd=1, qd=DNSQR(qname="google.com"))
[>>> reply = sr1(packet)
Begin emission
...
Finished sending 1 packets
*
Received 4 packets, got 1 answers, remaining 0 packets
[>>> █
```





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Conclusion	In this experiment, packet crafting and analysis were successfully performed using Scapy and Wireshark. Different types of packets such as ICMP and UDP were manually created and transmitted over the network. The ICMP packet generated a valid echo reply, demonstrating reliable communication and confirming the working of the request–response mechanism. In contrast, the UDP packet did not receive a response, which highlights the connectionless nature of UDP and its lack of guaranteed delivery or acknowledgment. All transmitted packets, including requests and responses, were successfully captured and analyzed in Wireshark, providing a clear understanding of packet structure and real-time network behavior. Overall, the experiment helped in understanding how network packets are constructed, transmitted, and analyzed, and provided practical insight into the working of different network protocols.
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